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POWDER MONITORING FOR ADDITIVE MANUFACTURING SYSTEMS

Abstract

An additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component, a powder delivery device configured to direct a powder stream toward the melt pool, at least one sensor configured to generate sensor data, and a computing device. The computing device may be configured to receive the sensor data from the at least one sensor, determine, based on the sensor data, at least one powder flow characteristic, and generate a signal indicative of the at least one powder flow characteristic. The computing device may be further configured to control, based on the at least one powder flow characteristic, the energy delivery device and the powder delivery device to deposit a plurality of layers based on a set of deposition parameters.

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Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to additive manufacturing techniques.

BACKGROUND

[0002] Additive manufacturing generates three-dimensional structures through addition of material layer-by-layer or volume-by-volume to form the structure, rather than removing material from an existing component to generate the three-dimensional structure. Additive manufacturing may be advantageous in many situations, such as rapid prototyping, forming components with complex three-dimensional structures, or the like. In some examples, additive manufacturing may utilize powdered materials and may melt or sinter the powdered material together in predetermined shapes to form the three-dimensional structures. In addition to initial fabrication, additive manufacturing techniques may include subsequent machining processes to finish a component.

SUMMARY

[0003] The disclosure describes additive manufacturing systems, and methods for operating additive manufacturing systems, that monitor powder flow. For example, at least one powder flow characteristic (e.g., powder velocity) may affect deposit quality and/or cause deviation of deposition parameters from nominal targets. However, determining powder velocity at one or more points along a powder flow path (e.g., in real-time or near real-time) presents challenges. In some examples, at least one sensor (e.g., an acoustic sensor, an in-line gas flow sensor, or a powder flow monitor), may be used to detect at least one powder flow characteristic. In this way, the additive manufacturing systems described herein may enable monitoring and control of powder flow in additive fabrication of a component. In turn, monitoring and control of powder flow may promote quality of production and reduce deviation of a product from a nominal product geometry.

[0004] In some examples, the disclosure describes an additive manufacturing system that includes an energy delivery device, a powder delivery device, at least one sensor, and a computing device. The energy delivery device is configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component. The powder delivery device is configured to direct a powder stream toward the melt pool. The at least one sensor is configured to generate sensor data. The computing device may be configured to receive the sensor data from the at least one sensor, determine, based on the sensor data, at least one powder flow characteristic, and generate a signal indicative of the at least one powder flow characteristic. The computing device may be further configured to control, based on the at least one powder flow characteristic, the energy delivery device and the powder delivery device to deposit a plurality of layers based on a set of deposition parameters.

[0005] In some examples, the disclosure describes a method that includes receiving, by a computing device, sensor data from at least one sensor of an additive manufacturing system. The method may further include determining, by the computing device, based on the sensor data, at least one powder flow characteristic. The method may further include generating, by the computing device, a signal indicative of the at least one powder flow characteristic. The method may further include controlling, by the computing device, based on the at least one powder flow characteristic, an energy delivery device and a powder delivery device of the additive manufacturing system to deposit a plurality of layers based on a set of deposition parameters.

[0006] The details of one or more examples are set forth in the accompanying drawings and the

description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. 1 is a conceptual block diagram illustrating powder monitoring aspects of an example additive manufacturing system configured to monitor powder flow of a powder used to fabricate a component during an additive manufacturing technique.

[0008] FIG. 2 is a conceptual and schematic diagram illustrating an example powder flow monitoring system configured to monitor powder flow between a powder delivery device and a build surface during the additive manufacturing technique.

[0009] FIG. 3 is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique.

[0010] FIG. 4 is a flowchart illustrating an example technique for fabricating a component while monitoring a powder flow characteristic.

DETAILED DESCRIPTION

[0011] The disclosure generally describes techniques and systems for monitoring at least one powder flow characteristic during a blown powder additive manufacturing technique (e.g., directed energy deposition, DED). For example, at least one powder flow characteristic (e.g., powder velocity) may be unknown at a given position in an additive manufacturing system and/or point of time during fabrication. Further, it may be difficult to control process parameters to achieve a predetermined powder flow characteristic without determining values of the powder flow characteristic during different process conditions or fabrication stages.

[0012] During blown powder additive manufacturing, a component is built up by adding material to the component in sequential layers, such that the final component is composed of a plurality of layers of material. In blown powder additive manufacturing techniques for forming components from metals or alloys, an energy source may direct energy at a substrate to form a melt pool, and a powder delivery device may deliver a powder to the melt pool. At least some of the powder at least partially melts and is joined to the melt pool and, thus, the substrate. Powder velocity influences deposit quality and the occurrence of adverse or abnormal events, for example, in flow of powder toward the substrate, or interaction of powder with the melt powder, or other events. For example, powder velocity may affect many process responses, such as build quality, build height, or layer thickness. Component geometry also influences powder flow (e.g., nozzle sizes), and powder flow beyond a nominal range may result in an abnormal interaction of powder with a component (for example, leading to blockage or plugging).

[0013] In accordance with techniques of the disclosure, an additive manufacturing system may include at least one sensor for detecting at least one powder flow characteristic. By monitoring at least one powder flow characteristic, the systems described herein may enable a more efficient use of powder in fabricating a component, more effectively and/or efficiently control operation of additive manufacturing. For example, at least one sensor may be used to monitor a velocity of a powder cloud in situ or ex situ. Powder velocity (or another powder flow characteristic) may be correlated to at least one deposit quality metric or an abnormal event (e.g., clogging, spattering, or plugging). Process parameters may be controlled (e.g., by adjusting powder flow rate, carrier gas pressure, or carrier gas flow rate) based on at least one powder flow characteristic detected by at least one sensor. Monitoring powder flow thus may promote product quality, control of additive fabrication within specifications, a better understanding of relation of process parameters and product quality, for selection of appropriate components or for using a wider range of components (e.g., a variety of a nozzle) to be used to achieve a predetermined powder velocity.

[0014] FIG. 1 is a conceptual block diagram illustrating an example additive manufacturing system **10** configured to monitor powder flow of a powder used to fabricate a component **22** during an additive manufacturing technique. In the example illustrated in FIG. 1, additive manufacturing system **10** includes a computing device **12**, a powder delivery device **14**, an energy delivery device **16**, a powder flow monitoring system (PFMS) **18**, a stage **20**, an optical system **50**, a melt pool monitoring system (MPMS) **52**, a powder source **42**, a powder source mass sensor **44**, and a sensor **48**.

[0015] Stage **20** is configured to position component **22** during an additive manufacturing process. In some examples, stage **20** is movable relative to energy delivery device **16** and/or energy delivery device **16** is movable relative to stage **20**. Similarly, stage **20** may be movable relative to powder delivery device **14** and/or powder delivery device **14** may be movable relative to stage **20**. Stage **20** may be configured to selectively position and restrain component **22** in place relative to stage **20** during manufacturing of component **22**.

[0016] Powder source **42** is the source of powder for powder stream **30**. Powder source **42** may include any suitable container or enclosure, such as a hopper, configured to hold powder. Powder source **42** also may include a mechanism for entraining the powder in a gas flow. For instance, powder source **42** may be coupled to a gas source, which provides a gas flowing through powder source **42** and entraining powder within the gas flow. Additionally, or alternatively, powder source **42** may include an agitator configured to agitate the powder and increase entrainment of the powder in the gas stream. System **10** may include a powder source mass sensor **44** associated with powder source **42**. Powder source mass sensor **44** may be configured to quantify loss of mass in the powder source **42** or, alternatively, a mass flow out of powder source **42**.

[0017] Powder source **42** is fluidically coupled to powder delivery device **14** via a flow path **46**. Flow path **46** may include any suitable structure(s) defining an enclosed flow between powder source **42** and powder delivery device, including conduit, pipe, tubes, or the like. Although not shown in FIG. 1, for at least part of flow path **46** between powder source **42** and nozzles of powder delivery device **14**, flow path **46** may split into multiple, parallel sections, e.g., one for each nozzle. Further, although not shown in FIG. 1, in some examples, flow path **46** may include one or more nozzles for controlling flow through flow path **46** as a whole or portions of flow path **46** (e.g., a section associated with a particular nozzle of powder delivery device **14**).

[0018] Powder delivery device **14** may be configured to deliver powder to selected locations of component **22** being formed via a powder stream **30**. Powder delivery device **14** may include one or more nozzles that each output powder, such that the combined powder defines powder stream **30** focused at a focus plane. As powder delivery device **14** is movable in the z-axis shown in FIG. 1 relative to component **22**, the focal plane of powder delivery device **14** also may be movable in the z-axis relative to component **22**, such that the focus plane may be controlled to be substantially coincident with a build surface **28** of component **22**.

[0019] In some examples, powder delivery device **14** may be mechanically coupled or attached to energy delivery device **16** to facilitate delivery of powder stream **30** and energy **34** for forming melt pool **32** to substantially the same location adjacent to component **22**. Energy delivery device **16** may include an energy source, such as a laser source, an electron beam source, plasma source, or another source of energy **34** that may be absorbed by component **22** to form a melt pool **32** and/or be absorbed by powder in powder stream **30** to be added to component **22**. Example laser sources include a CO laser, a CO.sub.2 laser, a Nd:YAG laser, or the like. In some examples, the energy source may be selected to provide energy with a predetermined wavelength or wavelength spectrum that may be absorbed by component **22** and/or the powder to be added to component **22** during the additive manufacturing technique.

[0020] In some examples, energy delivery device **16** also includes an energy delivery head, which is operatively connected to the energy source. The energy delivery head may aim, focus, or direct energy **34** toward predetermined positions at or adjacent to a surface of component **22** during the

additive manufacturing technique. As described above, in some examples, the energy delivery head may be movable in at least one dimension (e.g., translatable and/or rotatable) under control of computing device **12** to direct the energy toward a selected location at or adjacent to a surface of component **22**.

[0021] As shown in FIG. **1**, energy delivery device **16** may be arranged or configured such that energy **34** and powder stream **30** both exit from a common deposition head and are directed toward build surface **28**. For instance, energy **34** may pass through a central channel within the deposition head and exit a central aperture in the deposition head, while fluidized powder may flow through internal channels and powder nozzle(s) for forming powder stream **30** and directing powder stream **30** toward build surface **28**. At least some of the powder in powder stream **30** may impact a melt pool **32** in component **22**, and at least some of the powder that impacts melt pool **32** may be joined to component **22**.

[0022] During additive manufacturing, component **22** is built up by adding material to component **22** in sequential layers. The final component is composed of a plurality of layers of material. Energy delivery device **16** may direct energy **34** at first layer **24** to form melt pool **32**. Powder delivery device **14** may deliver powder stream **30** to melt pool **32**, where at least some of the powder at least partially melts and is joined to first layer **24**. Melt pool **32** cools as energy **34** is no longer delivered to that location of first layer **24** (e.g., due to energy delivery device **16** scanning energy **34** over the surface of first layer **24**). The temperature and cooling rate of melt pool **32** and the surrounding areas of first layer **24** affect the microstructure of the component **22** formed using the additive manufacturing technique.

[0023] System **10** may include both mass flow monitoring and heat flow monitoring. To provide mass flow monitoring, system **10** may include powder flow monitoring system (PFMS) **18**. PFMS **18** is configured to image at least a portion of powder stream **30** to detect powder flowing between powder delivery device **14** and build surface **28**. For example, PFMS **18** may include an illumination device and an imaging device. In some examples, the illumination device may include one or more light sources. For instance, the illumination device may include one or more structured light devices, such as one or more lasers. The illumination device is configured to illuminate a plane of powder stream **30** at image plane **38**, e.g., a plane substantially perpendicular to an axis extending between powder delivery device **14** and build surface **28**. The imaging device of PFMS **18** is configured to image at least some of the illuminated powder. The imaging device may have a relatively high data acquisition speed (e.g., frame rate), such as greater than 1000 Hz. To provide heat flow monitoring, system **10** may include melt pool monitoring system (MPMS) **52**. MPMS **52** is configured to image at least a portion of melt pool **32** to detect parameters, such as size, temperature, or shape, of melt pool **32**. For example, MPMS may be communicatively coupled to optical system **50** for observing thermal emissions around melt pool **32** and a thermal camera for monitoring a size and/or temperature of melt pool **32**.

[0024] Optical system **50** may include an imaging device and an associated optical train, which senses emissions at or near component **22** during the additive manufacturing technique. For example, optical system **50** may include a visible light imaging device, an infrared imaging device, or an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range. The optical train may include one or more reflective, refractive, diffractive optical components configured to direct light to the imaging device. For example, the optical train may be configured to direct light from near component **22** and/or melt pool **32** to the imaging device.

[0025] System **10** further includes at least one sensor **48**. Sensor **48** may be configured to generate sensor data. The sensor data may be representative of at least one powder flow characteristic of powder flowing through system **10**, in course of a process performed by system **10**. For example, the at least one powder flow characteristic may include at least one of a includes at least one of a powder flow rate, an in situ powder velocity, an ex situ powder velocity, a carrier gas flow rate, a

carrier gas pressure, a center purge flow rate, or a center purge pressure. Sensor **48** may include at least one of a powder flow monitor (e.g., a powder flow monitoring system described with reference to FIG. 2), an acoustic sensor, or an in-line gas flow sensor.

[0026] In some examples, sensor **48** includes at least one acoustic sensor. During operation of system **10**, an acoustic signal (e.g., sound) may be generated by flow of powder through system **10**, by interaction of powder with component **22** and/or melt pool **32**, or by interaction of power with some component of system **10**. The acoustic signal may be associated with one or more of a powder flow characteristic, a process parameter, or a condition of component **22** or a component of system **10**. In some examples, in which sensor **48** includes at least one acoustic sensor configured to generate at least one time-dependent acoustic data signal representative of the acoustic signal generated in or by system **10**.

[0027] The time-dependent acoustic data signal may be rich in information, and may include data representative of acoustic signals generated by powder flow, or by one or more component of system **10**. The acoustic signals may also be representative of process parameters or variations in process parameters. For example, the acoustic signal generated by powder delivery device **14** may depend upon process attributes including, for example, geometry of powder delivery device **14**, including wear of powder delivery device **14**, material deposit within powder delivery device **14**, powder flow rate through powder delivery device **14**, powder flow pulsing, carrier gas flow rate, carrier gas pressure within powder delivery device **14**, composition of powder, or the like.

Similarly, the acoustic signal generated by powder flow through different portions, regions, or components of system **10**, or toward melt pool **32**, or incorporating with melt pool **32**, may vary with variances in powder flow. In this way, computing device **12** may analyze the time-dependent acoustic data signal to determine whether powder flow and fabrication process parameters are within a nominal or expected range, or if the powder flow or fabrication process parameters varying compared to an expected value.

[0028] In some examples, sensor **48** includes a plurality of acoustic sensors, and respective acoustic sensors may be positioned near respective components of the system **10**. In some examples, each acoustic sensor of the plurality of acoustic sensors may generate a respective at least one time-dependent acoustic data signal. Because of the different positions of the respective acoustic sensors, the computing device may analyze the respective time-dependent acoustic data signals to determine information related to respective components of system **10**. For example, each respective time-dependent acoustic data signal may be associated with the respective component to which the respective acoustic sensor is near. Alternatively or additionally, the computing device may utilize the intensity of respective frequency components of at least one time-dependent acoustic data signal to determine, e.g., based on distance, to which component the sound may be attributed. In this way, the computing device may analyze the time-dependent acoustic data signal or time-dependent acoustic data signals to determine process attributes for a plurality of components of system **10**.

[0029] The at least one acoustic sensor may be positioned external to a housing that holds one or more components of system **10**, or may be positioned within the housing. The at least one acoustic sensor may include, for example, an acoustic sensing element such as a microphone or a sound-to-electric transducer or electromagnetic, capacitive, or piezoelectric elements that generate an electrical signal in response to incident sound waves.

[0030] The at least one acoustic sensor may be configured to sense acoustic signals with a predetermined wavelength or wavelength range. In some examples, the at least one acoustic sensor may be configured to sense acoustic signals that may or may not be detectable by human hearing, including infrasound and ultrasound. In some examples, the acoustic signals may include frequencies below about 20 Hz, from about 20 Hz to about 20 kHz, from about 20 kHz to about 2 MHz, higher than about 2 MHz, or combinations thereof. Each acoustic sensor of the at least one acoustic sensor is configured to generate a respective time-dependent acoustic data signal of at least

one time-dependent acoustic data signal based on the sensed acoustic signal, and communicate at least one time-dependent acoustic data signal to computing device **12**. In some examples, at least one time-dependent acoustic data signal includes a digital data signal, and at least one acoustic sensor includes an analog-to-digital converter. In other examples, at least one time-dependent acoustic data signal includes an analog signal. In some examples, the at least one acoustic sensor includes an amplifier to amplify the signal sensed by the at least one acoustic sensor and produce the at least one time-dependent acoustic data signal. The at least one acoustic sensor may transmit the at least one time-dependent acoustic data signal using electrical signals, Bluetooth, Wi-Fi, radio, or any other suitable transmission pathway.

[0031] In some examples, the at least one acoustic sensor may be configured to enhance detection of one acoustic signal compared to another acoustic signal. For instance, a first acoustic sensor of the at least one acoustic sensor may be positioned adjacent to a selected component of system **10**, oriented toward a selected component of system **10**, or the like to enhance detection of a selected acoustic signal compared to another acoustic signal. For example, a first acoustic sensor of the at least one acoustic sensor may be positioned to sense acoustic signals originating from powder delivery device **14**, and a second acoustic sensor of the at least one acoustic sensor may be positioned to sense acoustic signals originating from powder source **42**. The at least one acoustic sensor may be located near a component or at a zone within system **10**, or may be oriented towards a component to sense sound from the component, or otherwise more accurately attribute the sound to a source. In an example, the at least one acoustic sensor may include multiple acoustic sensors forming an acoustic sensor network that captures sound generated by various components of system **10**.

[0032] In some examples, sensor **48** includes at least one in-line gas flow sensor. The gas flow rate sensor may be configured to detect a flow rate of gas, for example, of gas flowing through powder delivery device **14**, along flow path **46**, or along some other component or conduit through which gas flows in system **10**. The flow rate may include a volumetric flow rate or a mass flow rate of the gas. The in-line gas flow sensor may include additional sensors configured to detect one or more parameters of the gas flow. Such parameters of the gas flow may include, but are not limited to, a velocity of the gas flow, a mass flow rate of the gas flow, a volumetric flow rate of the gas flow, a turbulence of the gas flow, a volume fraction of the gas flow with respect to the powder flow, a composition of the gas flow, a temperature of the gas flow, or any other parameter of the gas flow that may affect a material property of the as-deposited layer and/or may be modified through deposition parameters of powder delivery device **14** and/or energy delivery device **16**.

[0033] Computing device **12** is configured to control components and operation of system **10** and may include, for example, a desktop computer, a laptop computer, a workstation, a server, a mainframe, a cloud computing system, or the like. Computing device **12** may be communicatively coupled to, and configured to control, powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, sensor **48**, optical system **50**, and/or MPMS **52** using respective communication connections. Although FIG. **1** illustrates a single computing device **12** and attributes all control and processing functions to that single computing device **12**, in other examples, system **10** may include multiple computing devices **12**, e.g., a plurality of computing devices **12**.

[0034] Computing device **12** may be configured to control operation of powder delivery device **14**, energy delivery device **16**, stage **20**, and/or sensor **48** to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, sensor **48**, and/or MPMS **52**, during additive manufacturing processes. For example, as described above, computing device **12** may control stage **20** and powder delivery device **14**, energy delivery device **16**, and/or sensor **48** to translate and/or rotate along at least one axis to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, sensor **48**, and/or MPMS **52**. Positioning component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, sensor

48, and/or MPMS **52** may include positioning a predetermined surface (e.g., a surface to which material is to be added) of component **22** in a predetermined orientation relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, sensor **48**, and/or MPMS **52**.

[0035] Computing device **12** may be configured to control system **10** to deposit layers **24** and **26** to form component **22** based on a set of deposition parameters. The set of deposition parameters may include energy, feed, and motion parameters that are configured to produce layers **24**, **26**, having various physical parameters, such as a height of layers **24**, **26**, and a density of layers **24**, **26**. For example, the set of deposition parameters may include power, beam diameter, beam profile, and wavelength of energy delivery device **16**; powder feed rate and gas feed rate of powder delivery device **14**; scan speed and deposition path of stage **20** relative to energy delivery device **16** and powder delivery device **14**; or any other operating parameters that may affect an amount and/or quality of material formed as layers **24**, **26**.

[0036] Computing device **12** may be configured to select deposition parameters that are configured to generate layers **24**, **26**, according to predetermined physical parameters. As shown in FIG. **1**, component **22** may include a first layer **24** and a second layer **26**, although many components may be formed of additional layers, such as tens of layers, hundreds of layers, thousands of layers, or the like. Component **22** in FIG. **1** is simplified in geometry and the number of layers compared to many components formed using additive manufacturing techniques. Although techniques are described herein with respect to component **22** including first layer **24** and second layer **26**, the technique may be extended to components **22** with more complex geometry and any number of layers.

[0037] To form component **22**, computing device **12** may control powder delivery device **14** and energy delivery device **16** according to the set of operating parameters to form, on a surface **28** of first layer **24** of material, a second layer **26** of material. Computing device **12** may control energy delivery device **16** to deliver energy **34** to a volume at or near surface **28** to form melt pool **32**. For example, computing device **12** may control the relative position of energy delivery device **16** and stage **20** to direct energy to the volume. Computing device **12** also may control powder delivery device **14** to deliver powder stream **30** to melt pool **32**. For example, computing device **12** may control the relative position of powder delivery device **14** and stage **20** to direct powder stream **30** at or on to melt pool **32**.

[0038] Computing device **12** may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete. Computing device **12** may then control a z-axis position of stage **20** and/or powder delivery device **14** and energy delivery device **16** such that melt pool **32** will be formed on surface **36** of second layer **26**, and may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete.

[0039] Computing device **12** may use sensor data from sensor **48** (alone, or in combination with sensor data from at least one other sensor) to monitor at least one powder flow characteristic in system **10**, for example, to monitor a powder velocity. For example, computing device **12** may be configured to receive sensor data from sensor **48**. Computing device **12** may be further configured to determine, based on the sensor data, at least one powder flow characteristic, and generate a signal indicative of the at least one powder flow characteristic. The at least one powder flow characteristic may include any suitable characteristic that may be associated with powder flow, including, for example, at least one of a powder flow rate, an in situ powder velocity, an ex situ powder velocity, a carrier gas flow rate, a carrier gas pressure, a center purge flow rate, or a center purge pressure.

[0040] An operator or computing device **12** may perform an action in response to the signal indicative of the at least one powder flow characteristic. For example, computing device **12** may be further configured to control, based on the at least one powder flow characteristic, energy delivery

device **16** and/or powder delivery device **14** to deposit a plurality of layers based on a set of deposition parameters. Thus, the operator or computing device **12** may account for variances in powder flow in view of the deposition parameters to continue fabrication of component **22** (e.g., instead of interrupting or terminating fabrication). Alternatively, if powder flow deviates beyond acceptable thresholds, the operator or computing device **12** may interrupt or terminate fabrication, enabling further inspection of system **10** to assess possible causes of variances in powder flow. [0041] In some examples, computing device **12** is further configured to determine, based on the at least one powder flow characteristic, at least one deposit quality metric or at least one abnormal event. For example, the at least one deposit quality metric may be indicative of geometric conformance of component **22** to a model based on which component **22** is fabricated, or other quality indicators based on partially built component **22**, or of layers **24** or **26**. The at least one abnormal event may include accumulation or formation of material within or on a nozzle. Presence of such material may cause at least one of clogging of a powder orifice or of a gas orifice, spattering, distortion of the powder stream, formation of fluggies or plugging. Fluggies are a deposit or localized buildup of molten or semi-molten material on a surface of a nozzle. Presence of fluggies on or around a powder or gas stream orifice may produce interruptions or inconsistencies in the powder stream and the subsequent build. Fluggies may be caused by splatter or rebounding of semi-molten particles from or around the melt pool. In some cases fluggies will grow to an extent that may clog or cover some or most of the nozzle opening.

[0042] Computing device **12** may also use sensor data from sensor **48** to determine a quality of component **22**, for example, conformance of or deviation of component **22** compared to geometric tolerances. For example, computing device **12** may be configured to determine, based on the at least one powder flow characteristic, at least one process response. Computing device **12** may be further configured to determine, by comparing the process response with a threshold response value, the at least one quality metric. The at least one process response may be indicative of an aspect of component **22** influenced by a process performed by system **10**. For example, the at least one process response may include at least one of a build quality, a build height, or a layer thickness.

[0043] Computing device **12** may modify operation of system **10** in response to the at least one powder flow characteristic, or in response a parameter determined based on the at least one powder flow characteristic). For example, computing device **12** may be further configured to adjust, based on the at least one deposit quality metric or at least one abnormal event, at least one powder control parameter. The at least one powder control parameter may include at least one of a powder feed rate, a carrier gas flow rate, a carrier gas pressure, a center purge flow rate, a center purge pressure, or a nozzle standoff distance.

[0044] FIG. **2** is a conceptual and schematic diagram illustrating an example powder flow monitoring system (PFMS) **60** configured to monitor powder flow between a powder delivery device **62** and a build surface (not shown in FIG. **2**) during an additive manufacturing technique. Powder delivery device **62** may be an example of powder delivery device **14** of FIG. **1**, and PFMS **60** may be an example of PFMS **18** of FIG. **1**.

[0045] Powder delivery device **62** includes a deposition head **64** that carries a plurality of powder nozzles **66**. Plurality of powder nozzles **66** output a powder stream **68** toward the build surface. As shown in FIG. **2**, powder stream **68** may be focused at a focal plane, such that powder stream **68** is converging toward the focal plane and diverging away from the focal plane.

[0046] PFMS **60** includes a housing **70** (also referred to as an enclosure), which encloses an imaging device **72** and an illumination device **74**. In some examples, imaging device **72** may be a high-speed camera and illumination device **74** may be laser illuminator. Housing **70** is attached to an adjustable z-stage **76** by a bracket **78**.

[0047] Housing **70** is configured to enclose imaging device **72** and illumination device **74** and help protect imaging device **72** and illumination device **74** from a surrounding environment. For instance, housing **70** may be configured to surround imaging device **72** and illumination device **74**

and prevent any powder that reflects from the build surface toward PFMS **60** from impacting imaging device **72** or illumination device **74**.

[0048] Further, housing **70** may be configured to cool imaging device **72** and illumination device **74**. Imaging device **72** and illumination device **74** may be exposed to heat from the melt pool at the build surface and energy from the energy delivery device. Imaging device **72** and illumination device **74** may be relatively sensitive to heat and have improved operational lifetime if maintained and operated below a certain temperature. PFMS **60** may include a cooling system **80** configured to remove heat from within housing **70** to cool imaging device **72** and illumination device **74**. For instance, cooling system **80** may include cooling fluid circuit through which a cooling fluid flows, and housing **70** may include part of the cooling circuit. In some examples, housing **70** may be formed from a material having relatively high thermal conductivity, such as aluminum, to help transfer heat from within housing **70** to cooling system **80** (e.g., a cooling fluid flowing through cooling system **80**).

[0049] As described above, PFMS **60** may be configured to measure powder flow of powder stream **68** (FIG. 2) at one or more axial (or longitudinal) locations of powder stream **68** and determine one or more parameters associated with the powder flow. For instance, illumination device **74** may illuminate powder of powder stream **68** in a plane oriented substantially orthogonal to a longitudinal axis that extends from powder delivery device **62** to the build surface. PFMS **60** may be positioned at a selected axial or longitudinal location to image a selected axial or longitudinal position between powder delivery device **62** and the build surface. Imaging device **72** may be configured to image at least some of the illuminated powder.

[0050] System **10** may be configured with various in-situ monitoring techniques, including powder flow monitoring, mass flux monitoring, and heat flux monitoring, to control an additive manufacturing process.

[0051] FIG. 3 is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique. The technique of FIG. 3 may be implemented by system **10** of FIG. 1 or system **60** of FIG. 2 and will be described with concurrent reference to FIGS. 1 and 2. However, it will be appreciated that system **10** may perform other techniques and the technique of FIG. 3 may be performed by other systems.

[0052] Computing device **12** may be configured to monitor at least one powder flow characteristic, and control system **10** based on the at least one powder flow characteristic. For example, computing device **12** may be configured to control a powder feed rate output by powder source **42** (see top left of FIG. 3). For instance, computing device **12** may be configured to control an agitator of powder source **42**, a gas flow rate of gas flowing through powder source **42**, a position of one or more valves within flow path **46**, or the like to control a powder feed rate output by powder source **42**.

[0053] Computing device **12** may be configured to receive data from one or more mass flow monitoring sensors, including PFMS **18** and/or powder source mass sensor **44**. Data received from powder source mass sensor **44** indicates a mass flow of powder from powder source **42** to powder delivery device. Data from PFMS **18** indicates a mass flow of powder in powder stream **30** between powder delivery device **14** to adjacent melt pool **32**. In some examples computing device **12** may be configured to further receive sensor data from sensor **48**. For example, sensor **48** may include a powder flow sensor, and data from sensor **48** may indicate a topology of build surface **28**, and may further indicate powder mass captured by melt pool **32** and added to component **22**.

[0054] Computing device **12** may calculate one or more mass flow-related metrics based on the data received from PFMS **18**, powder source mass sensor **44**, and/or sensor **48**. For example, computing device **12** may determine a capture efficiency by determining a fraction or percentage of powder from powder stream **30** that is captured by melt pool **32** and added to component **22**, e.g., by dividing the powder mass captured by melt pool **32**, as determined based on data from powder flow sensor, into the mass flow determined based on data received from PFMS **18**. Further,

computing device **12** may determine an overall mass flux using the data received from PFMS **18**, powder source mass sensor **44**, and/or sensor **48**. Computing device **12** then may use the overall mass flux as an input to the control algorithm used to control the powder feed rate output by powder source **42** (see top left of FIG. **3**).

[0055] Similarly, computing device may be configured to control energy delivery device **16** to deliver energy **34** to first layer **24** to establish a given heat input (see bottom left of FIG. **3**). For example, one or more computing device **12** may control one or more operating parameters of energy delivery device **16**, such as intensity, pulse rate, pulse width, or the like; one or more positional parameters related to energy delivery device **16**, such as dwell time at a location, a movement rate relative to first layer **24**, an overlap between adjacent passes of energy **34** across first layer **24**, a pause time between adjacent passes of energy **34** across first layer **24**, or the like to control heat input to system **10** (e.g., to melt pool **32** and component **22**).

[0056] Computing device **12** may be configured to receive data from one or more heat sensors, such as optical system **50** and/or MPMS **52**. Computing device **12** may determine a cooling rate and associated heat from using data from optical system **50** and may determine a heat input into component using a size and/or temperature of melt pool **32** as observed by melt pool monitor. Computing device **12** may be configured to determine an overall heat flux using these data. Computing device **12** may then use the overall heat flux as an input to the control algorithm used to control the energy delivery by energy delivery device **16** (see bottom left of FIG. **3**). In some examples, computing device **12** may also use a deposit topology (captured powder mass) and/or capture efficiency metric in the determination of the heat flux, as the added powder mass and quench effects associated with the captured powder affect the cooling rate.

[0057] FIG. **4** is a flowchart illustrating an example technique for fabricating a component while monitoring a powder flow characteristic. The technique of FIG. **4** may be implemented by system **10** of FIG. **1** or system **60** of FIG. **2** and will be described with concurrent reference to FIGS. **1** and **2**. However, it will be appreciated that system **10** may perform other techniques and the technique of FIG. **3** may be performed by other systems.

[0058] In some examples, the technique of FIG. **4** includes receiving, by computing device **12**, sensor data from at least one sensor **48** of additive manufacturing system **10** (**100**). The sensor data is representative of at least one powder flow characteristic (associated with powder flowing through system **10**). At least one sensor **48** may include at least one of a powder flow monitor, an acoustic sensor, or an in-line gas flow sensor.

[0059] The technique may further include determining, by computing device **12**, based on the sensor data, at least one powder flow characteristic (**102**). The at least one powder flow characteristic may include at least one of a powder flow rate, an in situ powder velocity, an ex situ powder velocity, a carrier gas flow rate, a carrier gas pressure, a center purge flow rate, or a center purge pressure. Computing device **12** may use a predetermined correlation to determine the at least one powder flow characteristic based on the sensor data. The technique may further include generating, by computing device **12**, a signal indicative of the powder flow characteristic (**104**). Computing device **12** may generate an output indicative of the signal, or may transmit the signal to another component, device, or system. An operator, or computing device **12**, may take action in response to the signal. For example, the operator, or computing device **12**, may interrupt or terminate operation of system **10**, or modifying at least one process parameter of system **12**, in response to the signal.

[0060] The technique may further include, controlling, by computing device **12**, based on the at least one powder flow characteristic, energy delivery device **16** to deliver energy **34** to build surface **28** to form melt pool **32** and powder delivery device **14** to direct powder stream **30** toward melt pool **32** (**106**). Thus, computing device **12** may cause system **10** to deposit a plurality of layers based on a set of deposition parameters, accounting for the at least one powder flow characteristic. The controlling (**106**) may include, or be augmented by, determining other parameters of system

10. For example, the technique may further include determining, by computing device **12**, based on the at least one powder flow characteristic and at least one component interaction parameter, a mass flux, and controlling, based on the at least one powder control parameter and the mass flux, energy delivery device **16** and powder delivery device **14** to deposit the plurality of layers. The at least one component interaction parameter includes at least one of a part geometry, a melt pool capture capability, or a tool path. Further, the technique may include, by computing device **12**, at least one of: determining the part geometry based on a deposit topology; determining the melt pool capture capability based on melt pool size; or determining the tool path based on a build strategy. [0061] The technique may further include determining, by computing device **12**, at least one deposit quality metric or at least one abnormal event based on the at least one powder flow characteristic (**108**). The at least one abnormal event may include accumulation or formation of material within or on a nozzle, or any event indicating deviation of powder flow away from nominal flow characteristics. Presence of such material may cause at least one of clogging of a powder orifice or of a gas orifice, spattering, distortion of the powder stream, or plugging. Computing device **12** may assess a process response. For example, computing device **12** may determine at least one process response based on the at least one powder flow characteristic. The at least one process response may include at least one of a build quality, a build height, or a layer thickness. Computing device **12** may further determine the at least one quality metric by comparing the process response with a threshold response value. Thus, the quality metric may be indicative of conformance of component **22** to predetermined tolerances.

[0062] The technique may further include adjusting, by computing device **12**, at least one powder control parameter based on the at least one deposit quality metric or the at least one abnormal event (**110**). For example, the at least one powder control parameter includes at least one of a powder feed rate, a carrier gas flow rate, a carrier gas pressure, a center purge flow rate, a center purge pressure, or a nozzle standoff distance. Thus, computing device **12** may monitor and control system **10** while accounting for powder flow conditions in course of fabricating component **22**.

[0063] The following enumerated clauses describe various examples according to the present disclosure.

[0064] Clause 1: An additive manufacturing system, including: an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; at least one sensor configured to generate sensor data; and a computing device configured to: receive the sensor data from the at least sensor; determine, based on the sensor data, at least one powder flow characteristic; generate a signal indicative of the at least one powder flow characteristic; and control, based on the at least one powder flow characteristic, the energy delivery device and the powder delivery device to deposit a plurality of layers based on a set of deposition parameters.

[0065] Clause 2: The additive manufacturing system of clause 1, where the at least one sensor includes at least one of a powder flow monitor, an acoustic sensor, or an in-line gas flow sensor.

[0066] Clause 3: The additive manufacturing system of clauses 1 or 2, where the at least one powder flow characteristic includes at least one of a powder flow rate, an in situ powder velocity, an ex situ powder velocity, a carrier gas flow rate, a carrier gas pressure, a center purge flow rate, or a center purge pressure.

[0067] Clause 4: The additive manufacturing system of any of clauses 1 to 3, where the computing device is further configured to determine at least one deposit quality metric or at least one abnormal event based on the at least one powder flow characteristic.

[0068] Clause 5: The additive manufacturing system of clause 4, where the at least one abnormal event includes accumulation or formation of material within or on a nozzle.

[0069] Clause 6: The additive manufacturing system of clauses 4 or 5, where the computing device is further configured to determine at least one process response based on the at least one powder

flow characteristic.

[0070] Clause 7: The additive manufacturing system of clause 6, where the computing device is further configured to determine the at least one quality metric by comparing the process response with a threshold response value.

[0071] Clause 8: The additive manufacturing system of clauses 6 or 7, where the at least one process response includes at least one of a build quality, a build height, or a layer thickness.

[0072] Clause 9: The additive manufacturing system of any of clauses 4 to 8, where the computing device is further configured to adjust at least one powder control parameter based on the at least one deposit quality metric or the at least one abnormal event.

[0073] Clause 10: The additive manufacturing system of clause 9, where the at least one powder control parameter includes at least one of a powder feed rate, a carrier gas flow rate, a carrier gas pressure, a center purge flow rate, a center purge pressure, or a nozzle standoff distance.

[0074] Clause 11: A method for additive manufacturing, including: receiving, by a computing device, sensor data from at least one sensor of an additive manufacturing system; determining, by the computing device, based on the sensor data, at least one powder flow characteristic; generating, by the computing device, a signal indicative of the at least one powder flow characteristic; and controlling, by the computing device, based on the at least one powder flow characteristic, an energy delivery device and a powder delivery device of the additive manufacturing system to deposit a plurality of layers based on a set of deposition parameters.

[0075] Clause 12: The method of clause 11, where the at least one sensor includes at least one of a powder flow monitor, an acoustic sensor, or an in-line gas flow sensor.

[0076] Clause 13: The method of clauses 12 or 13, where the at least one powder flow characteristic includes at least one of a powder flow rate, an in situ powder velocity, an ex situ powder velocity, a carrier gas flow rate, a carrier gas pressure, a center purge flow rate, or a center purge pressure.

[0077] Clause 14: The method of any of clauses 11 to 13, further including determining, by the computing device, at least one deposit quality metric or at least one abnormal event based on the at least one powder flow characteristic.

[0078] Clause 15: The method of clause 14, where the at least one abnormal event includes accumulation or formation of material within or on a nozzle.

[0079] Clause 16: The method of clauses 14 or 15, further including determining, by the computing device, at least one process response based on the at least one powder flow characteristic.

[0080] Clause 17: The method of clause 15, further including determining, by the computing device, the at least one quality metric by comparing the process response with a threshold response value.

[0081] Clause 18: The method of clauses 16 or 17, where the at least one process response includes at least one of a build quality, a build height, or a layer thickness.

[0082] Clause 19: The method of any of clauses 14 to 18, further including adjusting, by the computing device, at least one powder control parameter based on the at least one deposit quality metric or the at least one abnormal event.

[0083] Clause 20: The method of clause 19, where the at least one powder control parameter includes at least one of a powder feed rate, a carrier gas flow rate, a carrier gas pressure, a center purge flow rate, a center purge pressure, or a nozzle standoff distance.

[0084] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware, or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with

other logic circuitry, or any other equivalent circuitry. A control unit including hardware may also perform one or more of the techniques of this disclosure.

[0085] Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various techniques described in this disclosure. In addition, any of the described units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware, firmware, or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware, firmware, or software components, or integrated within common or separate hardware, firmware, or software components.

[0086] The techniques described in this disclosure may also be embodied or encoded in an article of manufacture including a computer-readable storage medium encoded with instructions. Instructions embedded or encoded in an article of manufacture including a computer-readable storage medium encoded, may cause one or more programmable processors, or other processors, to implement one or more of the techniques described herein, such as when instructions included or encoded in the computer-readable storage medium are executed by the one or more processors. Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a compact disc ROM (CD-ROM), a floppy disk, a cassette, magnetic media, optical media, or other computer readable media. In some examples, an article of manufacture may include one or more computer-readable storage media.

[0087] In some examples, a computer-readable storage medium may include a non-transitory medium. The term “non-transitory” may indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g., in RAM or cache).

[0088] Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. An additive manufacturing system, comprising: an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; at least one sensor configured to generate sensor data; and a computing device configured to: receive the sensor data from the at least one sensor; determine, based on the sensor data, at least one powder flow characteristic; generate a signal indicative of the at least one powder flow characteristic; and control, based on the at least one powder flow characteristic, the energy delivery device and the powder delivery device to deposit a plurality of layers based on a set of deposition parameters.
2. The additive manufacturing system of claim 1, wherein the at least one sensor comprises at least one of a powder flow monitor, an acoustic sensor, or an in-line gas flow sensor.
3. The additive manufacturing system of claim 1, wherein the at least one powder flow characteristic comprises at least one of a powder flow rate, an in situ powder velocity, an ex situ powder velocity, a carrier gas flow rate, a carrier gas pressure, a center purge flow rate, or a center purge pressure.
4. The additive manufacturing system of claim 1, wherein the computing device is further configured to determine at least one deposit quality metric or at least one abnormal event based on the at least one powder flow characteristic.
5. The additive manufacturing system of claim 4, wherein the at least one abnormal event

comprises accumulation or formation of material within or on a nozzle.

6. The additive manufacturing system of claim 4, wherein the computing device is further configured to determine at least one process response based on the at least one powder flow characteristic.

7. The additive manufacturing system of claim 6, wherein the computing device is further configured to determine the at least one quality metric by comparing the process response with a threshold response value.

8. The additive manufacturing system of claim 6, wherein the at least one process response comprises at least one of a build quality, a build height, or a layer thickness.

9. The additive manufacturing system of claim 4, wherein the computing device is further configured to adjust at least one powder control parameter based on the at least one deposit quality metric or the at least one abnormal event.

10. The additive manufacturing system of claim 9, wherein the at least one powder control parameter comprises at least one of a powder feed rate, a carrier gas flow rate, a carrier gas pressure, a center purge flow rate, a center purge pressure, or a nozzle standoff distance.

11. A method for additive manufacturing, comprising: receiving, by a computing device, sensor data from at least one sensor of an additive manufacturing system; determining, by the computing device, based on the sensor data, at least one powder flow characteristic; generating, by the computing device, a signal indicative of the at least one powder flow characteristic; and controlling, by the computing device, based on the at least one powder flow characteristic, an energy delivery device and a powder delivery device of the additive manufacturing system to deposit a plurality of layers based on a set of deposition parameters.

12. The method of claim 11, wherein the at least one sensor comprises at least one of a powder flow monitor, an acoustic sensor, or an in-line gas flow sensor.

13. The method of claim 11, wherein the at least one powder flow characteristic comprises at least one of a powder flow rate, an in situ powder velocity, an ex situ powder velocity, a carrier gas flow rate, a carrier gas pressure, a center purge flow rate, or a center purge pressure.

14. The method of claim 11, further comprising determining, by the computing device, at least one deposit quality metric or at least one abnormal event based on the at least one powder flow characteristic.

15. The method of claim 14, wherein the at least one abnormal event comprises accumulation or formation of material within or on a nozzle.

16. The method of claim 14, further comprising determining, by the computing device, at least one process response based on the at least one powder flow characteristic.

17. The method of claim 16, further comprising determining, by the computing device, the at least one quality metric by comparing the process response with a threshold response value.

18. The method of claim 16, wherein the at least one process response comprises at least one of a build quality, a build height, or a layer thickness.

19. The method of claim 14, further comprising adjusting, by the computing device, at least one powder control parameter based on the at least one deposit quality metric or the at least one abnormal event.

20. The method of claim 19, wherein the at least one powder control parameter comprises at least one of a powder feed rate, a carrier gas flow rate, a carrier gas pressure, a center purge flow rate, a center purge pressure, or a powder standoff distance.
